Early and locally advanced non-small-cell lung cancer: ESMO Clinical Practice Guideline for diagnosis, treatment and follow-up

SUPPLEMENTARY MATERIAL

SECTION 1. STAGING

Clinical staging

[^{18}F]2-fluoro-2-deoxy-D-glucose (FDG)–positron emission tomography (PET), ideally coupled with a contrast-enhanced computed tomography (CT) scan, should minimally cover the chest and upper abdomen (including liver, kidneys and adrenal glands).[1] If there is no contrast-enhanced CT combined with the PET, a separate contrast-enhanced CT should be carried out. Central nervous system (CNS) imaging is not required in clinical stage I non-small-cell lung cancer (NSCLC) as the incidence of brain metastases is low (~1%).[2,3] Since the incidence of CNS metastases increases with advancing disease stage, CNS imaging should be considered in stage II NSCLC.[4] CNS imaging is required in all patients with clinical stage III NSCLC on FDG–PET, as ~15% of these patients have asymptomatic brain metastases and FDG–PET is not suitable to detect them.[5] CNS imaging by gadolinium-enhanced magnetic resonance imaging (MRI) is preferred as it is more sensitive for detecting brain metastases than contrast-enhanced CT.[6] If MRI is contraindicated or unavailable within a reasonable time period, contrast-enhanced CT can be used as an alternative.

In case of peripheral lesions, transthoracic percutaneous fine-needle aspiration and/or core biopsy under imaging guidance (typically CT but also navigational bronchoscopy) is useful. Needle biopsy is associated with a diagnostic accuracy of >88%, sensitivity of 90% and a false-negative rate of 22%. The most significant disadvantage of transthoracic needle biopsy is a procedural risk of pneumothorax, with incidence ranging from 17%-50%. In a randomised non-inferiority trial, navigational bronchoscopy was reported to be non-inferior to transthoracic needle biopsy in patients with peripheral pulmonary nodules with a diameter of 10-30 mm (diagnosis in 79% versus 74% of patients, respectively).[7] Additionally, navigational bronchoscopy resulted in fewer complications (pneumothorax rate 3% versus 28% with transthoracic needle biopsy). Systematic endosonography-guided sampling

[endoscopic ultrasound (EUS) or endobronchial ultrasound (EBUS)] is used for patients with abnormal (>10 mm short axis diameter or FDG uptake) mediastinal and/or hilar lymph nodes (LNs) on CT and/or PET. For peripheral tumours >3 cm without mediastinal involvement on CT or PET–CT, and in case of no FDG uptake by the primary tumour, mediastinal staging is still advised due to the risk of false-negative mediastinal PET assessment.[8] Based on the randomised non-inferiority MEDIAST trial, a mediastinoscopy may be omitted after negative and representative systematic EUS or EBUS node sampling.[9] The rate of occult N2 disease after upfront resection (8.8%) was non-inferior compared with mediastinoscopy before resection (7.7%),[9] although concerns have been raised about the non-inferiority margin of 8%.[10] The prevalence of nodal disease was ~10% in the upfront resection group versus 15% in the upfront mediastinoscopy group, suggesting an imbalance between the groups.[10] Nevertheless, 6.3% morbidity, 0.6% mortality and 10 days of treatment delay were prevented with upfront resection compared with upfront mediastinoscopy.[9] As a potential downside, mediastinal node relapses were more frequent in patients with upfront resection compared with the mediastinoscopy group [12% versus 6%, respectively; odds ratio 0.5, 95% confidence interval (CI) 0.2-1.1].[11] Although the study was not powered to evaluate disease-free survival (DFS) and overall survival (OS), no significant differences in these secondary endpoints were observed between the two groups [OS adjusted hazard ratio (HR) 0.8, 95% CI 0.5-1.3; DFS adjusted HR 0.9, 95% CI 0.6-1.4].[11] Patients who fulfil the criteria for mediastinal staging (**Supplementary Figure S1**) but are planned for radical radiotherapy (RT) rather than surgery should also undergo systematic EUS or EBUS nodal staging. In the single-arm SEISMIC study, 12% of patients had PET-occult LN metastases detected by EUS or EBUS, resulting in changes to treatment plans.[12]

Pathological staging

Accurate pathological staging using the ninth edition of the Union for International Cancer Control TNM (tumour–node–metastasis) Classification of Malignant Tumours[13] and the third edition of the International Association for the Study of Lung Cancer (IASLC) Staging Manual[14] is essential for treatment decision making, including perioperative or adjuvant therapy. EBUS or EUS fine-needle aspiration is the method of choice for sampling mediastinal LNs, ideally combined with rapid on-site cytological evaluation, if available.[15]

The definition of ‘adequate’ intraoperative LN sampling remains controversial, and the practical value of LN counts is questionable.[16,17] A station-based approach (e.g. ≥3 N2 and ≥3 N1 nodal stations) may be superior to the count-based approach (e.g. ≥10 LNs).[16] The interpretability of total LN count is hampered by the fact that pathologists may count fragmented LNs as unique LNs. Although extranodal extension in mediastinal LN metastases is often reported due to its association with poor prognosis, its utility as a future pathological N denominator remains to be determined.[16,18]

For part-solid non-mucinous adenocarcinoma, only the invasive (i.e. non-lepidic) tumour rather than the total size contributes to the T category, yet both must be recorded.[19] This downstaging rule does not apply to rare mucinous adenocarcinomas with a lepidic component.[20] Spread through air space is a pattern of tumour spread and is an adverse prognostic feature, but it is not included in tumour size for staging.[19]

Pathological staging on surgically resected samples is required for adjuvant therapy decisions. Pathological examination of samples, including LNs, should follow guidelines to allow application of the current pathological IASLC/TNM staging criteria.[14] Many pathological factors have been studied in early-stage lung cancer, such as grade and inflammation,[21,22] but there is currently insufficient evidence to support the use of prognostic features beyond IASLC/TNM stage to inform adjuvant therapy decisions.

If multiple tumours share the same histological subtype (e.g. adenocarcinoma or squamous-cell carcinoma), next-generation sequencing analysis is more reliable than morphological assessment to distinguish between multiple individual primaries and intrapulmonary metastasis.[23,24] The requirement for pathological subtyping and molecular analysis will depend on available resources and treatment options.

Pathological response assessment is required to measure treatment effectiveness of various preoperative regimens including chemotherapy (ChT), immunotherapy or a combination of both. Major pathological response (defined as ≤10% viable tumour cells) and pathological complete response (no viable tumour cells) are considered potential surrogate prognostic endpoints.[25] Recent recommendations provide

guidance for standardised processing of lung resection specimens and histological assessment of pathological response.[26] Pathological stage reporting after preoperative systemic therapy should follow IASLC/TNM rules and use the yp prefix. ypT stage is calculated by the proportional reduction of the maximum tumour bed diameter according to the percentage of residual viable tumour; for example, a 60 mm mass (T3) with 10% residual viable tumour becomes ypT1a (6 mm viable tumour).

SECTION 2. PRE-TREATMENT RISK ASSESSMENT

All patients with NSCLC should be evaluated in a tumour board setting by a multidisciplinary team (MDT) comprising surgeons, medical oncologists, radiation oncologists, pathologists, pulmonologists, radiologists and supportive care staff. A complete medical history with comorbidities, weight loss, performance status (PS) and physical examination must be recorded. Smoking history must be included, indicating type, quantity and timing. As surgery and RT are potentially curative treatments for localised disease, maximum consideration should be given to whether these treatments are feasible. Factors such as smoking-related or age-related comorbidities may impact treatment decisions. As any locoregional therapy will impair pulmonary and vascular reserve to some extent, tolerability of the therapeutic intervention should be evaluated carefully. Management of lung cancer in the setting of interstitial lung disease (ILD) is a common clinical scenario, as ~5%-10% of patients with lung cancer have a concurrent diagnosis of ILD.[27] Patients with ILD have greater rates of post-operative mortality (5.1% versus 1.2% in those without ILD), acute respiratory distress syndrome (1.9% versus 0.5%) and composite morbidity and mortality (13.2% versus 7.4%).[28]

As immunotherapy is increasingly incorporated in the treatment regimen, attention should be paid to the use of immunosuppressive drugs, presence of autoimmune disorders (including previous and current treatments) and the presence of ILD.

Standard laboratory tests, including routine haematology and tests of renal and hepatic function, are required. Baseline testing of thyroid function and adrenal function is encouraged in patients planned for immunotherapy. Patients planned for

adjuvant osimertinib require an electrocardiogram and assessment of cardiac function.

In general, patients with unresectable stage II-III disease, a World Health Organization PS score of 0-1 and no major comorbidities are candidates for definitive chemoradiotherapy (CRT) if they fulfil RT planning constraints in line with European guidelines.[29]

Pre-treatment surgical risk

For surgical candidates, algorithms for pre-treatment evaluation have been developed and are used widely (**Supplementary Figure S2**).[30,31] Cardiac risk assessment for lung resection should follow the recalibrated thoracic revised cardiac risk index (ThRCRI).[32] In an external validation cohort, the incidence of cardiac morbidity in patients with risk index scores of 0, 1-1.5, 2-2.5 and >2.5 were 0.9%, 4.2%, 8% and 18%, respectively.[33] Patients with ThRCRI score ≥2, a pre-existing cardiac condition or limited exercise tolerance should be referred for cardiac consultation and managed in line with the American College of Cardiology/American Heart Association guideline.[34]

The relative risk of post-operative morbidity and mortality can be predicted from preoperative forced expiratory volume in the first second (FEV_1) and the diffusing capacity of the lung for carbon monoxide (DLCO). Predicted post-operative (ppo) values can be calculated by multiplying the preoperative values by the number of segments expected to remain after surgery and dividing by the number of preoperative functioning segments. When pneumonectomy is planned, a quantitative radionuclide perfusion scan is useful to estimate the contribution of the lung to be resected. Surgical resection is acceptable with a low risk of morbidity and mortality if ppo-FEV_1 and ppo-DLCO values are ≥60%; however, if they are <60%, additional testing is needed (**Supplementary Figure S2**). A large database analysis from 2009-2011 reported that the incidence of complications and mortality was associated with ppo-FEV_1 and ppo-DLCO.[35] The study also showed that outcomes after video-assisted thoracoscopic surgery (VATS) were more favourable than after thoracotomy. When ppo-FEV_1 or ppo-DLCO was <40%, operative mortality by open procedure was 4.8% and 5.2%, respectively, following open surgery, and 0.7% and 2.0%,

respectively, following VATS.[35] In patients with substantial emphysema, however, ppo-FEV_1 is not significantly reduced and may even increase when undergoing lobectomy (lung volume reduction surgery effect). Thus, ppo-FEV_1 is not a reliable predictor of post-operative complications in patients with chronic obstructive pulmonary disease (COPD).[36] The improvement of FEV_1 after surgery is observed when the COPD index (defined as FEV_1%/100 + FEV_1/forced vital capacity) is <1.0.[37] Patients with severe COPD should not be denied potentially curative surgery based on guidelines established in patients without COPD.

Patients with lower values (ppo-FEV_1 or ppo-DLCO <30%) should undergo cardiopulmonary exercise testing (CPET). When maximal oxygen consumption (VO_2 max) is <10 ml/kg/min, patients are at high risk for serious post-operative complications.[30] CPET is not, however, available at every institution. Substitutes for CPET include the shuttle walk test and the stair climbing test. Although there is a lack of standardisation of these tests, they may be useful as a preliminary evaluation of patients' cardiopulmonary function.[30]

Operability should be jointly assessed by thoracic surgeons and pulmonologists. In addition to CPET, imaging such as VQ scanning may help to evaluate regional emphysema and inform operative planning, particularly in patients who may benefit from lung volume reduction effects. Ischaemia testing can be useful in patients with cardiac risk factors (e.g abnormal electrocardiogram, ThRCRI score ≥2) to guide preoperative optimisation.

SECTION 3. STEREOTACTIC BODY RT TOXICITY

Stereotactic body RT (SBRT) is very well tolerated, even in elderly and/or frail patients. High-grade toxicity, deterioration of quality of life and deterioration of pulmonary function are uncommon.[38] Patients with interstitial pulmonary fibrosis are at highest risk for severe pulmonary toxicity. In a prospective study in which patient inclusion was carried out after central review by a dedicated ILD team and SBRT delivery techniques were optimised, SBRT (50 Gy in five fractions) led to fatal pneumonitis in 7.7% of patients, which is less than half the rate reported in previous retrospective series.[39] Median OS was 25 months.

Fatal bleeding was observed in a study of SBRT (60-66 Gy in three fractions) in patients with central or ultracentrally located tumours.[40] Prospective studies have reported high rates of fatal haemoptysis as a late effect when the main blood vessels have received RT doses higher than 80 Gy equivalent dose in 2 Gy.[41-44] Risk factors for bleeding include tumour invasion into the airways or blood vessels and the use of anticoagulants. A recent study used more fractionated schedules for non-peripheral tumours, with fewer, but still significant, late side-effects.[42] In a recent systematic review, hypofractionated RT in 8-18 fractions was considered appropriate for ultracentral lesions near the proximal tracheobronchial tree, upper trachea and oesophagus.[45] For other ultracentral lesions near the heart, great vessels, brachial plexus and spine, or for non-ultracentral but still central lesions, five-fraction SBRT was considered an appropriate option, while adhering to dose constraints applied to organs at risk in recent trials. If SBRT is not feasible due to dose constraints to organs at risk, 'classic' fractionation is an alternative. In a small randomised controlled trial, 70 Gy in 35 fractions led to similar progression-free survival and OS compared with SBRT, with a slightly higher rate of low-grade pneumonitis and oesophagitis.[46]

SECTION 4. SUPERIOR SULCUS TUMOURS

Superior sulcus tumours are potentially curable after multimodality therapy to achieve resection with no tumour at the margin (R0). Patients should be assessed by a surgical team with expertise in complex surgery following induction therapy. These tumours pose a surgical challenge due to the morbidity associated with attempts to achieve R0 resection. While prospective randomised data are lacking, expert opinion supports the use of neoadjuvant CRT in patients with T3-4 N0-1 superior sulcus tumours. The largest prospective study in this population (single-arm SWOG 9146) used two cycles of cisplatin–etoposide with concurrent RT at 45 Gy (1.8 Gy per fraction);[47] however, higher doses of RT are acceptable (45-60 Gy; 1.8-2.0 Gy per fraction). Resectability should be discussed by the MDT before and after treatment. A remaining mass, even when FDG-avid, does not imply that the tumour is unresectable. After surgery, adjuvant targeted therapy or immunotherapy can be utilised according to biomarker testing results (see sections 'Adjuvant targeted therapy' and 'Adjuvant ICI therapy').

It is unclear how many patients with superior sulcus tumours were included in recent neoadjuvant or perioperative trials (apart from KEYNOTE-671, which explicitly excluded them) but their number was likely very small due to the rarity of the disease. Given the location of the tumour, which makes it difficult to obtain clear resection margins without damaging adjacent vital structures, concurrent CRT followed by surgery and adjuvant chemoimmunotherapy remains the recommended approach, as stated in the latest National Comprehensive Cancer Network guidelines.[48] Nonetheless, including such patients in future neoadjuvant trials could potentially improve their outcomes.

In patients with tumours deemed unresectable after CRT, durvalumab consolidation is an accepted option (see section 'Management of unresectable stage III NSCLC'). The role of neoadjuvant or perioperative ChT combined with immune checkpoint inhibitors and resection without RT for these tumours is unclear.

SECTION 5. SURVEILLANCE

Clinical and radiographical surveillance after curative-intent treatment aims to improve patient outcomes by detecting locoregional recurrences that may be amenable to curative salvage therapy, metastatic recurrences that are amenable to systemic therapy while the patient remains well, and second primaries at an early stage for curative therapy if possible. The IFCT-0302 study randomised patients with resected stage I-IIIA NSCLC to undergo either CT-based surveillance every 6 months for 2 years and then annually for 5 years, or chest radiographical-based surveillance.[49] The study did not demonstrate any significant improvement in OS with CT-based surveillance, but CT enabled detection of more cases of early recurrence and second primary lung cancer. IFCT-0302 was carried out between 2005 and 2012 and thus the advent of more effective systemic therapies (e.g. immunotherapy, targeted therapy) may change patient outcomes with earlier detection of metastatic recurrence. The MinDose CT study demonstrated that minimal-dose CT was more sensitive for detecting recurrence of new cancers than chest X-ray (sensitivity 94% versus 21%, respectively; $P < 0.0001$) and had a higher negative predictive value (99% versus 96%, respectively; $P = 0.007$).[50]

The highest risk period for recurrence of NSCLC treated with curative-intent surgery

or CRT is within the first 2 years, with the potential for late recurrences extending to 5 years. The risk of developing a second primary lung cancer also remains elevated in this population. Follow-up visits at a minimum of every 6 months for the first 2 years and annually thereafter until 5 years can help to detect recurrences. Individual patient risk factors may necessitate more frequent monitoring.

Patients treated with RT as the primary modality require unique considerations during the surveillance period. Patients who receive SBRT may develop delayed local recurrence, which may be masked by treatment-related fibrosis. Potential high-risk features of changes in the RT field should be carefully monitored.[51,52] FDG–PET can help differentiate radiation-related fibrosis from recurrence.[53]

Smoking cessation is critical in all patients with lung cancer undergoing curative treatment, as this leads to superior treatment outcomes.[54]

SUPPLEMENTARY TABLES

Supplementary Table S1. Clinical classification of early and locally advanced NSCLC according to the UICC TNM ninth edition[13]

T (primary tumour)	
TX	Primary tumour cannot be assessed, or tumour proven by the presence of malignant cells in sputum or bronchial washings but not visualised by imaging or bronchoscopy
T0	No evidence of primary tumour
Tis	Carcinoma *in situ*[a]
T1	Tumour ≤3 cm in greatest dimension, surrounded by lung or visceral pleura, or tumour in a lobar or more peripheral bronchus[b]
T1mi	Minimally invasive adenocarcinoma[c]
T1a	Tumour ≤1 cm in greatest dimension[b]
T1b	Tumour >1 cm but ≤2 cm in greatest dimension[b]
T1c	Tumour >2 cm but ≤3 cm in greatest dimension[b]
T2a	Tumour >3 cm but ≤4 cm in greatest dimension **or** Tumour ≤4 cm in greatest dimension that invades ≥1 of the following: • The main bronchus (up to but not including the carina) • Visceral pleura • An adjacent lobe • Is associated with atelectasis or obstructive pneumonitis extending to the hilar region, involving either part of or the entire lung
T2b	Tumour >4 cm but ≤5 cm in greatest dimension

T3	Tumour >5 cm but ≤7 cm in greatest dimension **or** Tumour ≤7 cm that directly invades any of the following: • Parietal pleura, chest wall (including superior sulcus tumours), pericardium • Phrenic nerve, azygos vein,[d] thoracic nerve roots (i.e. T1, T2), stellate ganglion **or** • Separate tumour nodule(s) in the same lobe as the primary
T4	Tumour >7 cm **or** Tumour of any size that invades any of the following: • Mediastinum, thymus, trachea, carina • Recurrent laryngeal nerve, vagus nerve • Oesophagus, diaphragm • Heart, aorta, superior or inferior vena cava, intrapericardial pulmonary arteries or veins, supra-aortic arteries, brachiocephalic veins, subclavian vessels • Vertebral body, lamina, spinal canal, cervical nerve roots, brachial plexus **or** • Separate tumour nodule(s) in a different ipsilateral lobe to that of the primary
N (regional LNs)	
NX	Regional LNs cannot be assessed
N0	No regional LN metastasis
N1	Metastasis in ipsilateral peribronchial and/or ipsilateral hilar LNs and/or intrapulmonary LNs, including involvement by direct extension
N2	Metastasis in ipsilateral mediastinal and/or subcarinal LN station(s)
N2a	Metastasis in a single station

N2b	Metastasis in multiple ipsilateral stations
N3	Metastasis in contralateral mediastinal, contralateral hilar, ipsilateral or contralateral scalene or supraclavicular LN(s)
M (distant metastasis)	
M0	No distant metastasis
M1	Distant metastasis
M1a	Separate tumour nodule(s) in a contralateral lobe, tumour with pleural or pericardial nodules or malignant pleural or pericardial effusion[e]
M1b	Single extrathoracic metastasis in a single organ or organ system[f]
M1c	Multiple extrathoracic metastases in a single or multiple organ(s) or organ system(s)
M1c1	Multiple extrathoracic metastases in a single organ or organ system
M1c2	Multiple extrathoracic metastases in multiple organs or organ system(s)

Stage grouping			
Occult carcinoma	TX	N0	M0
Stage 0	Tis	N0	M0
Stage IA1	T1mi T1a	N0 N0	M0 M0
Stage IA2	T1b	N0	M0
Stage IA3	T1c	N0	M0
Stage IB	T2a	N0	M0

Stage IIA	T1	N1	M0
	T2b	N0	M0
Stage IIB	T1	N2a	M0
	T2	N1	M0
	T3	N0	M0
Stage IIIA	T1	N2b	M0
	T2	N2a	M0
	T3	N1	M0
	T3	N2a	M0
	T4	N0	M0
	T4	N1	M0
Stage IIIB	T1	N3	M0
	T2	N2b	M0
	T2	N3	M0
	T3	N2b	M0
	T4	N2a-b	M0
Stage IIIC	T3	N3	M0
	T4	N3	M0

LN, lymph node; NSCLC, non-small-cell lung cancer; TNM, tumour–node–metastasis; UICC, Union for International Cancer Control.

[a]Tis includes adenocarcinoma *in situ* and squamous-cell carcinoma *in situ*.

[b]The uncommon superficial spreading tumour of any size with its invasive component limited to the bronchial wall, which may extend proximal to the main bronchus, is also classified as T1a.

[c]Solitary adenocarcinoma (≤3 cm in greatest dimension) with a predominantly lepidic pattern and ≤5 mm invasion in greatest dimension in any one focus. Invasive mucinous adenocarcinomas should be staged as other invasive adenocarcinomas, i.e. only the invasive size being used as the T descriptor (regardless of the extent of the lepidic component).

[d]Although these structures lie within the mediastinum, invasion of these structures is not considered sufficient mediastinal penetration to be classified as a T4 category.

[e]Most pleural (pericardial) effusions with lung cancer are due to the tumour. In a few patients, however, multiple microscopic examinations of pleural (pericardial) fluid are negative for tumour, and the fluid is non-bloody and is not an exudate. Where these elements and clinical judgement dictate that the effusion is not related to the tumour, the effusion should be excluded as a staging descriptor.

[f]This includes the involvement of a single non-regional node.

Reprinted with permission.[13]

Supplementary Table S2. Levels of evidence and grades of recommendation (adapted from the Infectious Diseases Society of America-United States Public Health Service Grading System[a])

Levels of evidence

I	Evidence from at least one large randomised, controlled trial of good methodological quality (low potential for bias) or meta-analyses of well-conducted randomised trials without heterogeneity
II	Small randomised trials or large randomised trials with a suspicion of bias (lower methodological quality) or meta-analyses of such trials or of trials with demonstrated heterogeneity
III	Prospective cohort studies
IV	Retrospective cohort studies or case-control studies
V	Studies without control group, case reports, expert opinions

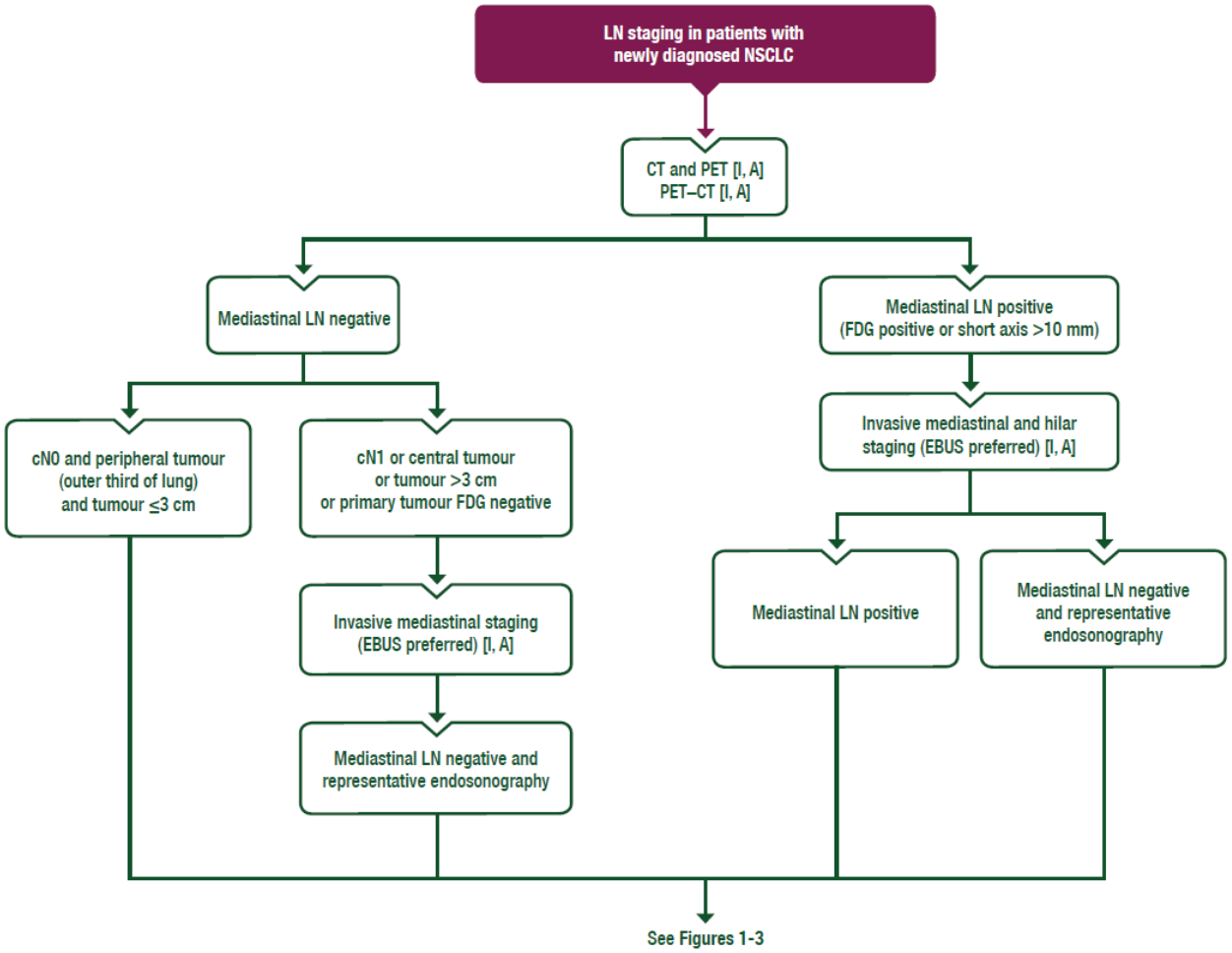
Grades of recommendation

A	Strong evidence for efficacy with a substantial clinical benefit, strongly recommended
B	Strong or moderate evidence for efficacy but with a limited clinical benefit, generally recommended
C	Insufficient evidence for efficacy or benefit does not outweigh the risk or the disadvantages (adverse events, costs, etc.), optional
D	Moderate evidence against efficacy or for adverse outcome, generally not recommended
E	Strong evidence against efficacy or for adverse outcome, never recommended

[a]By permission of Oxford University Press on behalf of the Infectious Diseases Society of America.[55]

SUPPLEMENTARY FIGURES

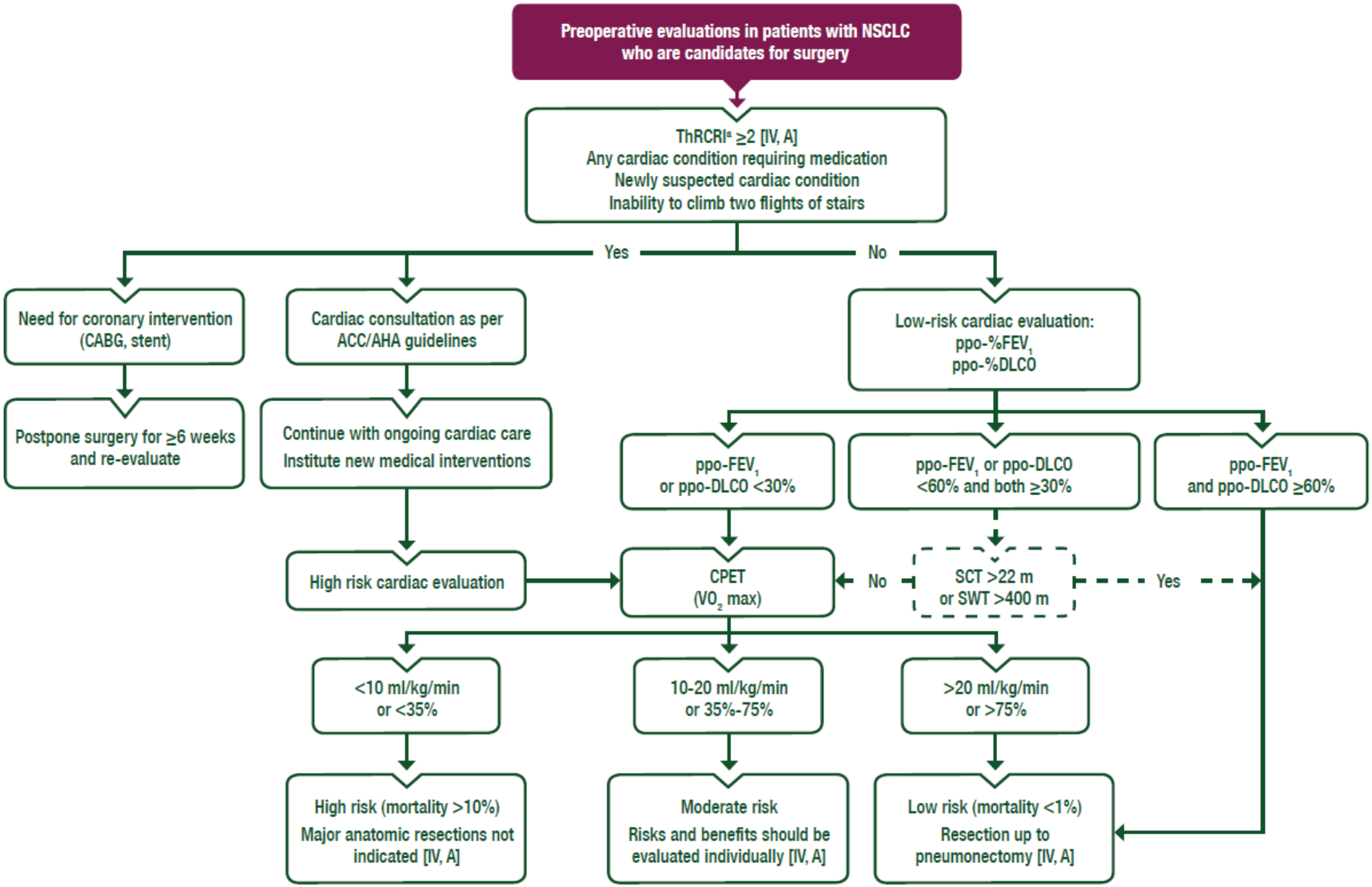
Supplementary Figure S1. LN staging in NSCLC.



Purple: algorithm title; white: other aspects of management.

c, clinical; CT, computed tomography; EBUS, endobronchial ultrasound; FDG, [^{18}F]2-fluoro-2-deoxy-D-glucose; LN, lymph node; NSCLC, non-small-cell lung cancer; PET, positron emission tomography.

Supplementary Figure S2. Preoperative cardiac and respiratory evaluation in candidates for lung resection.[30,31]



Purple: algorithm title; white: other aspects of management; dashed lines: optional branches, colour used as described in the categories above.

ACC, American College of Cardiology; AHA, American Heart Association; CABG, coronary artery bypass graft; CPET, cardiopulmonary exercise testing; DLCO, diffusing lung capacity for carbon monoxide; FEV_1, forced expiratory volume in the first second; NSCLC, non-small-cell lung cancer; ppo, predicted post-operative; SCT, stair climbing test; SWT, shuttle walk test; ThRCRI, thoracic revised cardiac risk index; TIA, transient ischaemic attack; VO_2 max, maximal oxygen consumption.

[a]Previous stroke or TIA = 1.5 points; previous ischaemic heart disease = 1.5 points; serum creatinine ≥2 mg/dl = 1 point; pneumonectomy = 1 point.

Reproduced with permission of the © ERS 2025; reprinted from Brunelli A, Varela G, Salati M, et al. Recalibration of the revised cardiac risk index in lung resection candidates. *Ann Thorac Surg*. 2010;90(1):199-203, with permission from Elsevier.[30,31]

REFERENCES

1. Gregory DL, Hicks RJ, Hogg A, et al. Effect of PET/CT on management of patients with non-small cell lung cancer: results of a prospective study with 5-year survival data. *J Nucl Med.* 2012;53(7):1007-1015.
2. Azenha LF, Bertoglio P, Kestenholz P, et al. Role of Pre-Operative Brain Imaging in Patients with NSCLC Stage I: A Retrospective, Multicenter Analysis. *Cancers (Basel).* 2022;14(10):2419.
3. Balekian AA, Fisher JM, Gould MK. Brain Imaging for Staging of Patients With Clinical Stage IA Non-small Cell Lung Cancer in the National Lung Screening Trial: Adherence With Recommendations From the Choosing Wisely Campaign. *Chest.* 2016;149(4):943-950.
4. Kim K, Lee J, Lee JY, et al. Clinical features and molecular genetics associated with brain metastasis in suspected early-stage non-small cell lung cancer. *Front Oncol.* 2023;13:1148475.
5. Hendriks LE, Bootsma GP, de Ruysscher DK, et al. Screening for brain metastases in patients with stage III non-small cell lung cancer: Is there additive value of magnetic resonance imaging above a contrast-enhanced computed tomography of the brain? *Lung Cancer.* 2013;80(3):293-297.
6. Schoenmaekers J, Hofman P, Bootsma G, et al. Screening for brain metastases in patients with stage III non-small-cell lung cancer, magnetic resonance imaging or computed tomography? A prospective study. *Eur J Cancer.* 2019;115:88-96.
7. Lentz RJ, Frederick-Dyer K, Planz VB, et al. Navigational Bronchoscopy or Transthoracic Needle Biopsy for Lung Nodules. *N Engl J Med.* 2025;392(21):2100-2112.
8. Vilmann P, Clementsen PF, Colella S, et al. Combined endobronchial and esophageal endosonography for the diagnosis and staging of lung cancer: European Society of Gastrointestinal Endoscopy (ESGE) Guideline, in cooperation with the European Respiratory Society (ERS) and the European Society of Thoracic Surgeons (ESTS). *Endoscopy.* 2015;47(6):545-559.
9. Bousema JE, Dijkgraaf MGW, van der Heijden E, et al. Endosonography With or Without Confirmatory Mediastinoscopy for Resectable Lung Cancer: A Randomized Clinical Trial. *J Clin Oncol.* 2023;41(22):3805-3815.

10. Farjah F. Tradeoffs of Proceeding to Resection Rather Than Mediastinoscopy After Negative Endosonography. *J Clin Oncol.* 2023;41(22):3782-3784.
11. Bousema JE, Spaans LN, Dijkgraaf MGW, et al. Two-year survival and disease recurrence after endosonography with or without confirmatory mediastinoscopy for resectable lung cancer (a short communication of the MEDIASTrial follow-up). *Lung Cancer.* 2025;202:108462.
12. Steinfort DP, Kothari G, Wallace N, et al. Systematic endoscopic staging of mediastinum to guide radiotherapy planning in patients with locally advanced non-small-cell lung cancer (SEISMIC): an international, multicentre, single-arm, clinical trial. *Lancet Respir Med.* 2024;12(6):467-475.
13. Union for International Cancer Control. *TNM Classification of Malignant Tumours.* 9th ed. Oxford, UK: John Wiley & Sons, Ltd; 2025.
14. International Association for the Study of Lung Cancer. Staging Manual in Thoracic Oncology, Third Edition. Available at https://www.iaslc.org/research-education/publications-resources-guidelines/staging-manual-thoracic-oncology-3rd-edition. Published 2024. Accessed February 24, 2025.
15. Sehgal IS, Dhooria S, Aggarwal AN, et al. Impact of Rapid On-Site Cytological Evaluation (ROSE) on the Diagnostic Yield of Transbronchial Needle Aspiration During Mediastinal Lymph Node Sampling: Systematic Review and Meta-Analysis. *Chest.* 2018;153(4):929-938.
16. Heiden BT, Eaton DB, Jr., Chang SH, et al. Assessment of Updated Commission on Cancer Guidelines for Intraoperative Lymph Node Sampling in Early Stage NSCLC. *J Thorac Oncol.* 2022;17(11):1287-1296.
17. Osarogiagbon RU, Van Schil P, Giroux DJ, et al. The International Association for the Study of Lung Cancer Lung Cancer Staging Project: Overview of Challenges and Opportunities in Revising the Nodal Classification of Lung Cancer. *J Thorac Oncol.* 2023;18(4):410-418.
18. Luchini C, Veronese N, Nottegar A, et al. Extranodal extension of nodal metastases is a poor prognostic moderator in non-small cell lung cancer: a meta-analysis. *Virchows Arch.* 2018;472(6):939-947.
19. Nicholson AG, Tsao MS, Beasley MB, et al. The 2021 WHO Classification of Lung Tumors: Impact of Advances Since 2015. *J Thorac Oncol.* 2022;17(3):362-387.

20. Travis WD, Asamura H, Bankier AA, et al. The IASLC Lung Cancer Staging Project: Proposals for Coding T Categories for Subsolid Nodules and Assessment of Tumor Size in Part-Solid Tumors in the Forthcoming Eighth Edition of the TNM Classification of Lung Cancer. *J Thorac Oncol.* 2016;11(8):1204-1223.
21. Zhu CQ, Pintilie M, John T, et al. Understanding prognostic gene expression signatures in lung cancer. *Clin Lung Cancer.* 2009;10(5):331-340.
22. Moreira AL, Ocampo PSS, Xia Y, et al. A Grading System for Invasive Pulmonary Adenocarcinoma: A Proposal From the International Association for the Study of Lung Cancer Pathology Committee. *J Thorac Oncol.* 2020;15(10):1599-1610.
23. Dacic S, Cao X, Bota-Rabassedas N, et al. Genomic Staging of Multifocal Lung Squamous Cell Carcinomas Is Independent of the Comprehensive Morphologic Assessment. *J Thorac Oncol.* 2024;19(2):273-284.
24. Nicholson AG, Torkko K, Viola P, et al. Interobserver Variation among Pathologists and Refinement of Criteria in Distinguishing Separate Primary Tumors from Intrapulmonary Metastases in Lung. *J Thorac Oncol.* 2018;13(2):205-217.
25. Deutsch JS, Cimino-Mathews A, Thompson E, et al. Association between pathologic response and survival after neoadjuvant therapy in lung cancer. *Nat Med.* 2024;30(1):218-228.
26. Travis WD, Dacic S, Wistuba I, et al. IASLC Multidisciplinary Recommendations for Pathologic Assessment of Lung Cancer Resection Specimens After Neoadjuvant Therapy. *J Thorac Oncol.* 2020;15(5):709-740.
27. Goodman CD, Nijman SFM, Senan S, et al. A Primer on Interstitial Lung Disease and Thoracic Radiation. *J Thorac Oncol.* 2020;15(6):902-913.
28. Axtell AL, David EA, Block MI, et al. Association Between Interstitial Lung Disease and Outcomes After Lung Cancer Resection. *Ann Thorac Surg.* 2023;116(3):533-541.
29. De Ruysscher D, Faivre-Finn C, Moeller D, et al. European Organization for Research and Treatment of Cancer (EORTC) recommendations for planning and delivery of high-dose, high precision radiotherapy for lung cancer. *Radiother Oncol.* 2017;124(1):1-10.

30. Brunelli A, Kim AW, Berger KI, et al. Physiologic evaluation of the patient with lung cancer being considered for resectional surgery: Diagnosis and management of lung cancer, 3rd ed: American College of Chest Physicians evidence-based clinical practice guidelines. *Chest.* 2013;143(suppl 5):e166S-e190S.
31. Brunelli A, Charloux A, Bolliger CT, et al. ERS/ESTS clinical guidelines on fitness for radical therapy in lung cancer patients (surgery and chemo-radiotherapy). *Eur Respir J.* 2009;34(1):17-41.
32. Brunelli A, Varela G, Salati M, et al. Recalibration of the revised cardiac risk index in lung resection candidates. *Ann Thorac Surg.* 2010;90(1):199-203.
33. Brunelli A, Cassivi SD, Fibla J, et al. External validation of the recalibrated thoracic revised cardiac risk index for predicting the risk of major cardiac complications after lung resection. *Ann Thorac Surg.* 2011;92(2):445-448.
34. Fleisher LA, Fleischmann KE, Auerbach AD, et al. 2014 ACC/AHA guideline on perioperative cardiovascular evaluation and management of patients undergoing noncardiac surgery: a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines. *Circulation.* 2014;130(24):e278-333.
35. Burt BM, Kosinski AS, Shrager JB, et al. Thoracoscopic lobectomy is associated with acceptable morbidity and mortality in patients with predicted postoperative forced expiratory volume in 1 second or diffusing capacity for carbon monoxide less than 40% of normal. *J Thorac Cardiovasc Surg.* 2014;148(1):19-28.
36. Yacoub WN, Meyers BF. Surgical resection in combination with lung volume reduction surgery for stage I non-small cell lung cancer. *Semin Thorac Cardiovasc Surg.* 2010;22(1):38-43.
37. Korst RJ, Ginsberg RJ, Ailawadi M, et al. Lobectomy improves ventilatory function in selected patients with severe COPD. *Ann Thorac Surg.* 1998;66(3):898-902.
38. Videtic GMM, Donington J, Giuliani M, et al. Stereotactic body radiation therapy for early-stage non-small cell lung cancer: Executive Summary of an ASTRO Evidence-Based Guideline. *Pract Radiat Oncol.* 2017;7(5):295-301.

39. Palma DA, Bahig H, Hope A, et al. Stereotactic Radiation Therapy in Early Non-Small Cell Lung Cancer and Interstitial Lung Disease: A Nonrandomized Clinical Trial. *JAMA Oncol.* 2024;10(5):575-582.
40. Timmerman R, McGarry R, Yiannoutsos C, et al. Excessive toxicity when treating central tumors in a phase II study of stereotactic body radiation therapy for medically inoperable early-stage lung cancer. *J Clin Oncol.* 2006;24(30):4833-4839.
41. Bezjak A, Paulus R, Gaspar LE, et al. Safety and Efficacy of a Five-Fraction Stereotactic Body Radiotherapy Schedule for Centrally Located Non-Small-Cell Lung Cancer: NRG Oncology/RTOG 0813 Trial. *J Clin Oncol.* 2019;37(15):1316-1325.
42. Lindberg K, Grozman V, Karlsson K, et al. The HILUS-Trial-a Prospective Nordic Multicenter Phase 2 Study of Ultracentral Lung Tumors Treated With Stereotactic Body Radiotherapy. *J Thorac Oncol.* 2021;16(7):1200-1210.
43. Levy A, Adebahr S, Hurkmans C, et al. Stereotactic Body Radiotherapy for Centrally Located Inoperable Early-Stage NSCLC: EORTC 22113–08113 LungTech Phase II Trial Results. *J Thorac Oncol.* 2024;19(9):1297-1309.
44. Swaminath A, Parpia S, Wierzbicki M, et al. Stereotactic vs Hypofractionated Radiotherapy for Inoperable Stage I Non-Small Cell Lung Cancer: The LUSTRE Phase 3 Randomized Clinical Trial. *JAMA Oncol.* 2024;10(11):1571-1575.
45. Yan M, Louie AV, Kotecha R, et al. Stereotactic body radiotherapy for Ultra-Central lung Tumors: A systematic review and Meta-Analysis and International Stereotactic Radiosurgery Society practice guidelines. *Lung Cancer.* 2023;182:107281.
46. Nyman J, Hallqvist A, Lund JA, et al. SPACE - A randomized study of SBRT vs conventional fractionated radiotherapy in medically inoperable stage I NSCLC. *Radiother Oncol.* 2016;121(1):1-8.
47. Rusch VW, Giroux DJ, Kraut MJ, et al. Induction chemoradiation and surgical resection for non-small cell lung carcinomas of the superior sulcus: Initial results of Southwest Oncology Group Trial 9416 (Intergroup Trial 0160). *J Thorac Cardiovasc Surg.* 2001;121(3):472-483.

48. National Comprehensive Cancer Network. Non-Small Cell Lung Cancer. Version 7.2025. Available at https://www.nccn.org/guidelines/guidelines-detail?category=1&id=1450. Published 2025. Accessed July 16, 2025.
49. Westeel V, Foucher P, Scherpereel A, et al. Chest CT scan plus x-ray versus chest x-ray for the follow-up of completely resected non-small-cell lung cancer (IFCT-0302): a multicentre, open-label, randomised, phase 3 trial. *Lancet Oncol.* 2022;23(9):1180-1188.
50. Hanna WC, Paul NS, Darling GE, et al. Minimal-dose computed tomography is superior to chest x-ray for the follow-up and treatment of patients with resected lung cancer. *J Thorac Cardiovasc Surg.* 2014;147(1):30-33.
51. Giuliani ME, Hope A, Mangona V, et al. Predictors and Patterns of Regional Recurrence Following Lung SBRT: A Report From the Elekta Lung Research Group. *Clin Lung Cancer.* 2017;18(2):162-168.
52. Spratt DE, Wu AJ, Adeseye V, et al. Recurrence Patterns and Second Primary Lung Cancers After Stereotactic Body Radiation Therapy for Early-Stage Non-Small-Cell Lung Cancer: Implications for Surveillance. *Clin Lung Cancer.* 2016;17(3):177-183.e172.
53. Lewis S, Chan M, Weiss J, et al. 3'-Deoxy-3'-(18F) Fluorothymidine Positron Emission Tomography/Computed Tomography in Non-small Cell Lung Cancer Treated With Stereotactic Body Radiation Therapy: A Pilot Study. *Adv Radiat Oncol.* 2022;7(6):101037.
54. Park ER, Gareen IF, Japuntich S, et al. Primary Care Provider-Delivered Smoking Cessation Interventions and Smoking Cessation Among Participants in the National Lung Screening Trial. *JAMA Intern Med.* 2015;175(9):1509-1516.
55. Dykewicz CA. Summary of the guidelines for preventing opportunistic infections among hematopoietic stem cell transplant recipients. *Clin Infect Dis*. 2001;33(2):139-144 [adapted from: Gross PA, Barrett TL, Dellinger EP, et al. Purpose of quality standards for infectious diseases. *Clin Infect Dis*. 1994;18(3):421].